

Atrial Fibrillation with Rapid Ventricular Response

Patient: *Robert Mitchell, 72*

Presentation

History of hypertension. Complains of palpitations and shortness of breath.

- HR: 152 bpm (irregularly irregular)
- BP: 108/68

Monitor

No P waves. Irregular QRS intervals.

Electrical Macro View

Multiple atrial ectopic foci overriding SA node.

Cellular Mechanism

Chaotic atrial myocyte depolarization → inconsistent AV conduction.

EMS Medication

Diltiazem and metoprolol

Why It Works

Blocks L-type Ca^{2+} channels in AV node → slows conduction → reduces ventricular rate.

If Unstable?

Synchronized cardioversion.